Question ID 67c0200a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 67c0200a

The number a is 70% less than the positive number b. The number c is 80% greater than a. The number c is how many times b?

ID: 67c0200a Answer

Correct Answer:
.54, 27/50

Rationale

The correct answer is $.54$. It's given that the number a is 70% less than the positive number b. Therefore, $a = \left(1 - \frac{70}{100}\right)b$, which is equivalent to $a = (1 - 0.70)b$, or $a = 0.30b$. It's also given that the number c is 80% greater than a. Therefore, $c = \left(1 + \frac{80}{100}\right)a$, which is equivalent to $c = (1 + 0.80)a$, or $c = 1.80a$. Since $a = 0.30b$, substituting $0.30b$ for a in the equation $c = 1.80a$ yields $c = 1.80(0.30b)$, or $c = 0.54b$. Thus, c is 0.54 times b. Note that .54 and 27/50 are examples of ways to enter a correct answer.

Question Difficulty:
Hard